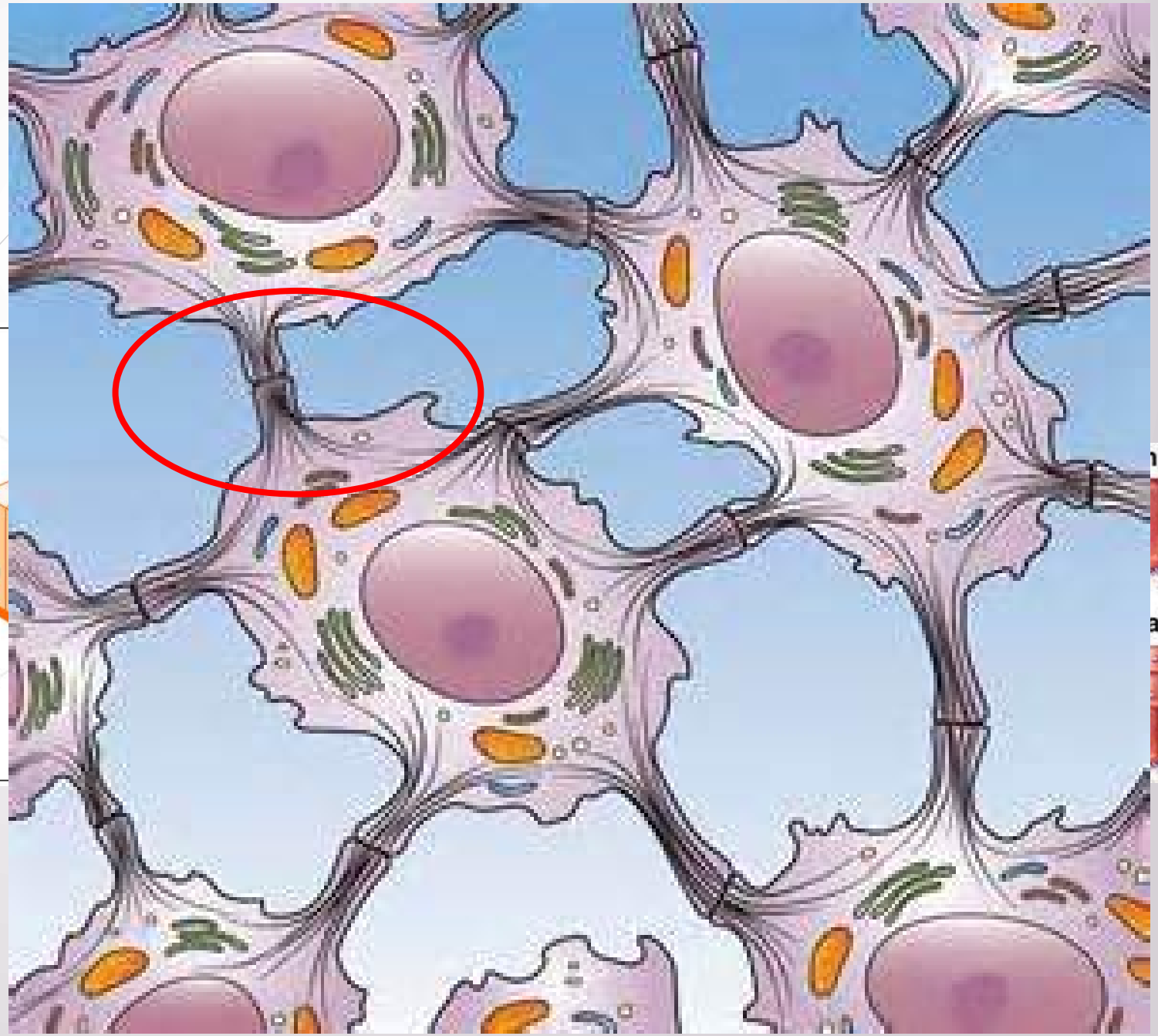
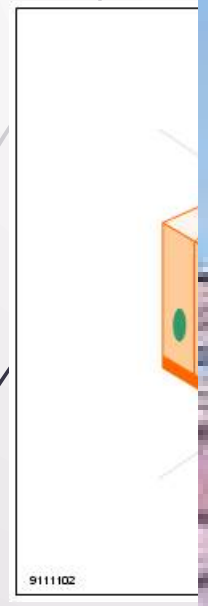
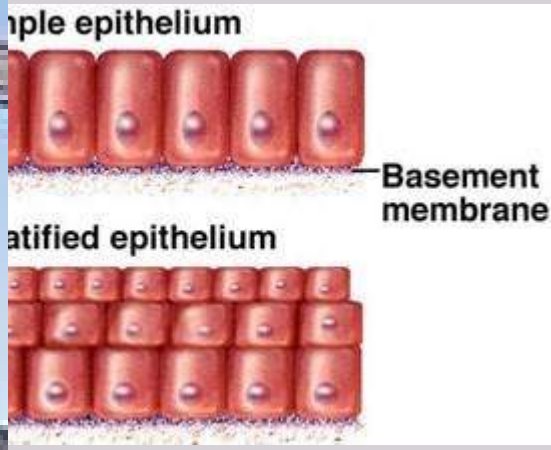
# CYTOSKELETON, CYTOSKELETAL JUNCTIONS FIBROBLASTS

Dr.FARZEEN SHAFIQ.




9111102


nple epithelium
Basement membrane
atified epithelium

# Cytoskeleton

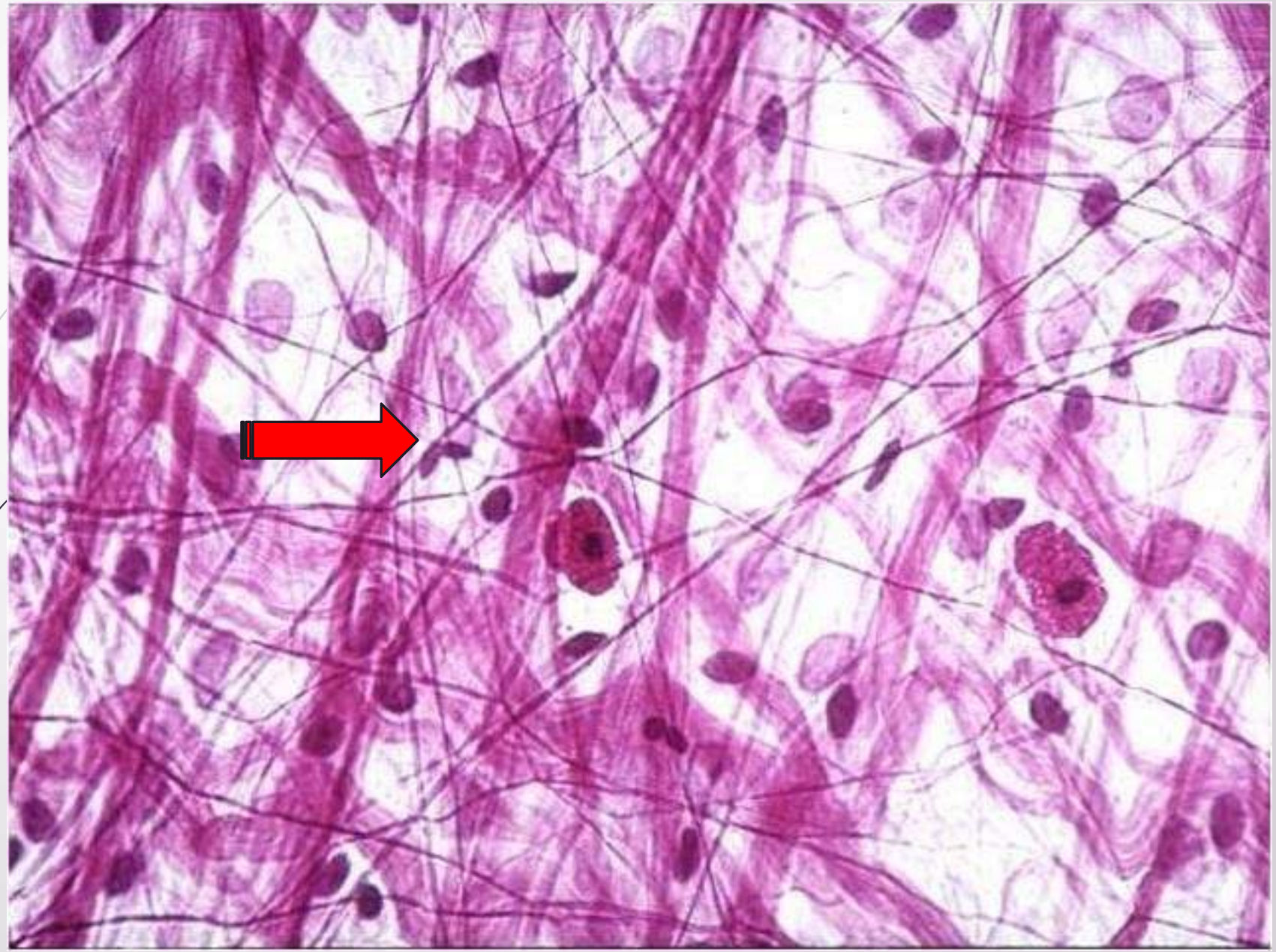
- Cells possess a cytoskeleton that provides a structural framework,

facilitates intracellular transport, supports cell junctions

and transmits signals about cell contact and adhesion,

and permits motility.

- The three structural elements of the cytoskeleton are *microfilaments, intermediate filaments,* and *microtubules*.
- All are dynamic structures assembled from protein subunits and disassembled as cellular activities and external influences on the cell change.



# MICROFILAMENTS

- *Microfilaments* are 6 to 8 nm in diameter and consist

of globular actin molecules.

- Microfilaments form tracks for the movement of myosin and
- serve as intracellular “muscles "for maintenance of cell shape, movement, and contractility

# *Intermediate filaments*

- are approximately 10 nm in diameter

and have a diverse protein composition.

- They are notcontractile but are important in the maintenance of cell

shape and contact between adjacent cells and the extracellular

matrix.

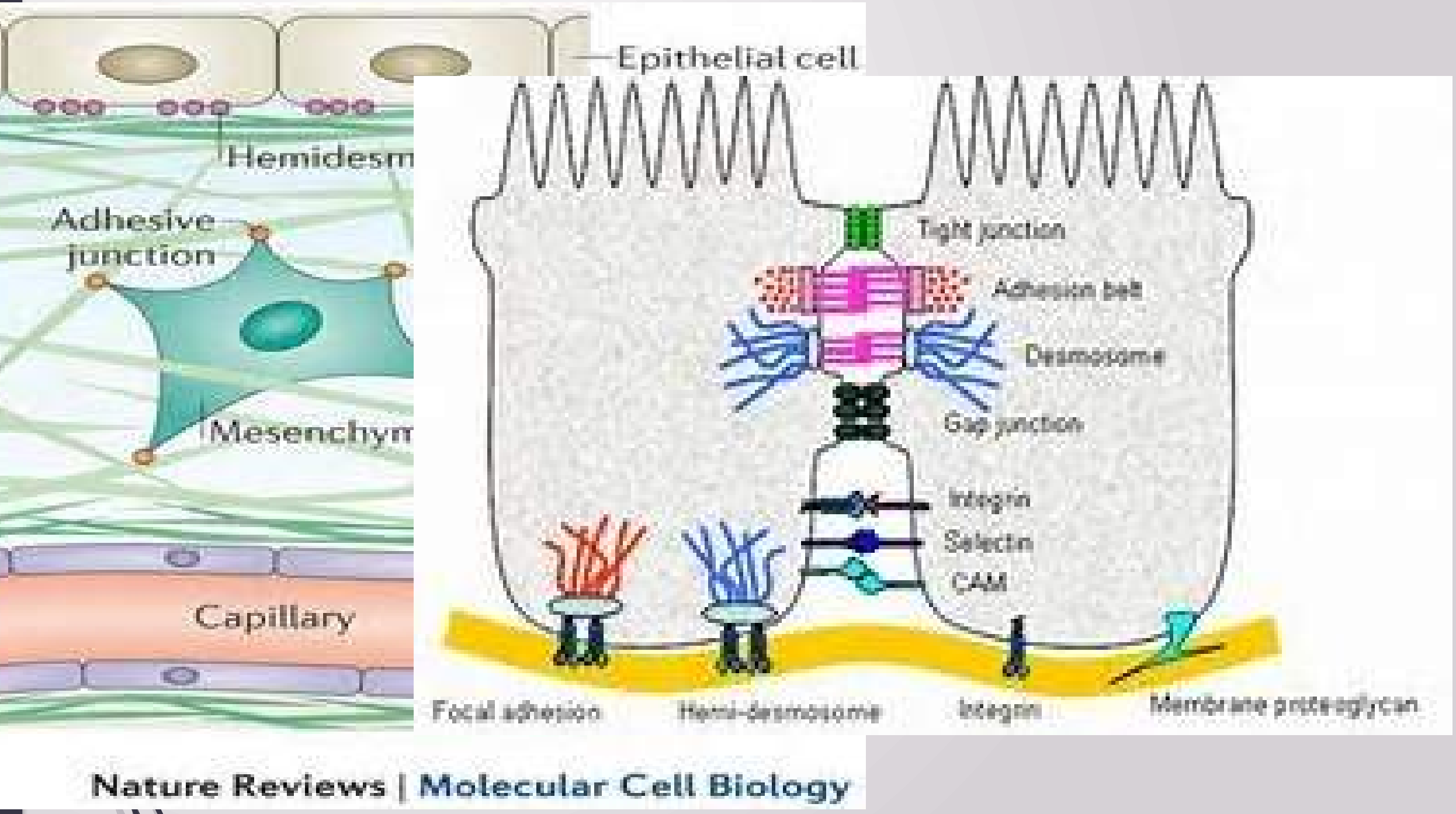
- In cells of mesenchymal origin, such as fibroblasts

and osteoblasts .

- In epithelial cells the filaments form bundles, called *tonofilaments,* which anchor onto desmosomes

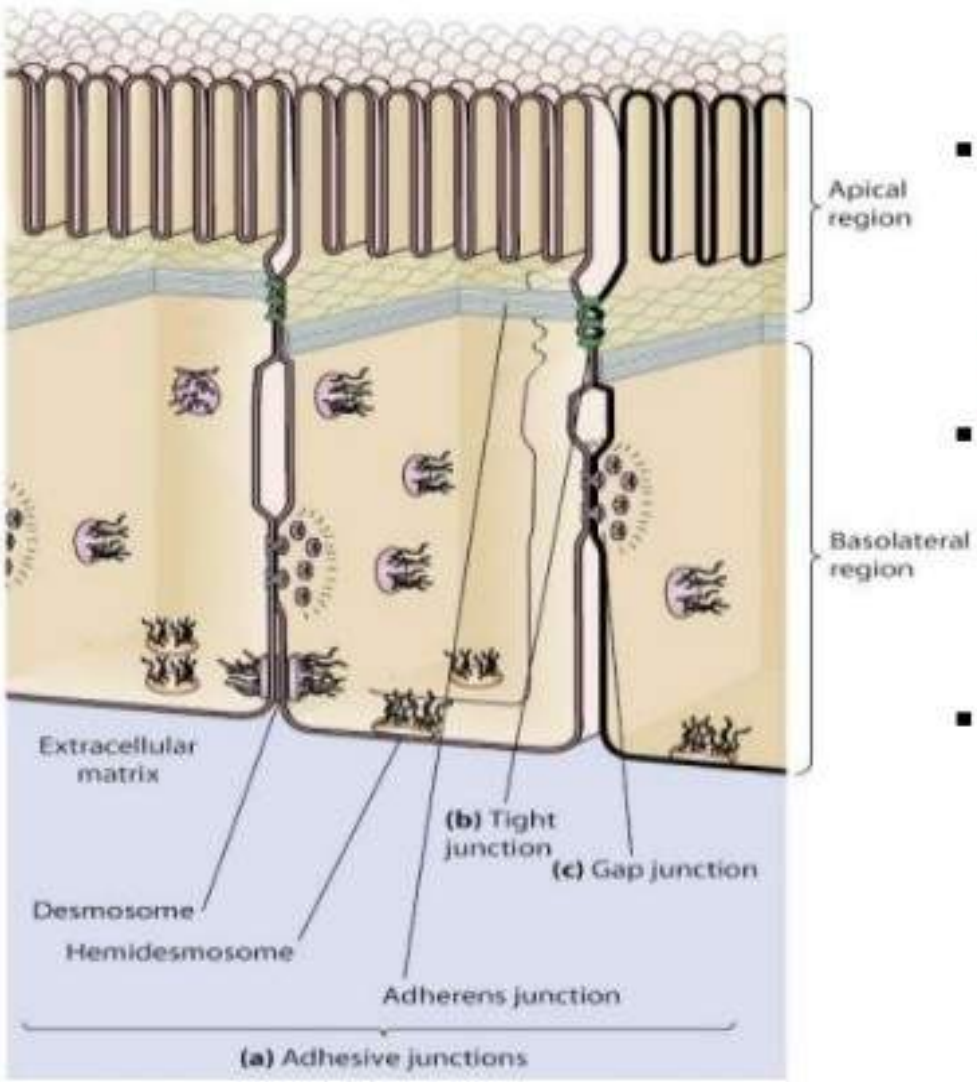
# MICROTUBULES

- *Microtubules* are tubular or cylindrical structures with an
- average diameter of 25 nm (Figure 4-3).
- Microtubules are composed of the protein *tubulin* arranged in rings stacked
- end to end, making up the tubules.
- Microtubules provide internal support for the cell; are the basis of motility for certain organelles, such as cilia; act as guide paths and part of the motor mechanism for the movement of secretory vesicles and other organelles; and serve to position certain organelles within the cell.

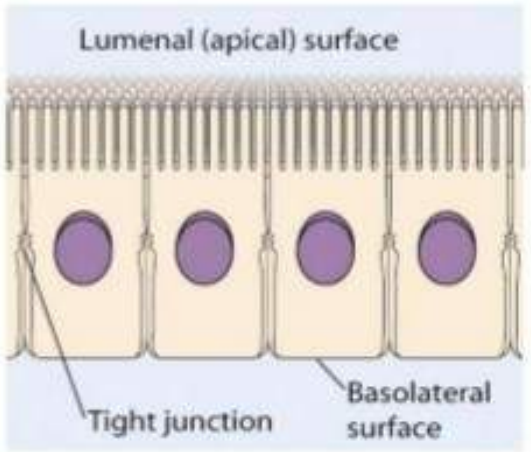



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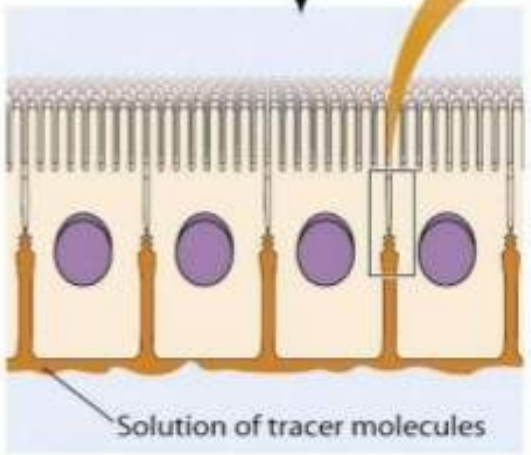
# CELL JUNCTIONS




- Cell junctions are the structures where long term association between neighboring cells are established.
- The 3 most common kinds of cell junctions are **tight junctions, adhesive/anchoring junctions,** and **gap/ communicating junctions.**
- Adhesive junctions (desmosomes, hemidesmosomes and adherens junctions) link adjoining cells to each other and to the ECM.
- Although adhesive junction types are similar in structure and function, they contain distinct intracellular attachment proteins and transmembrane linker proteins.




Electron-opaque tracer added to one side of cell layer




(a)



(b)

0.2 $\mu$m

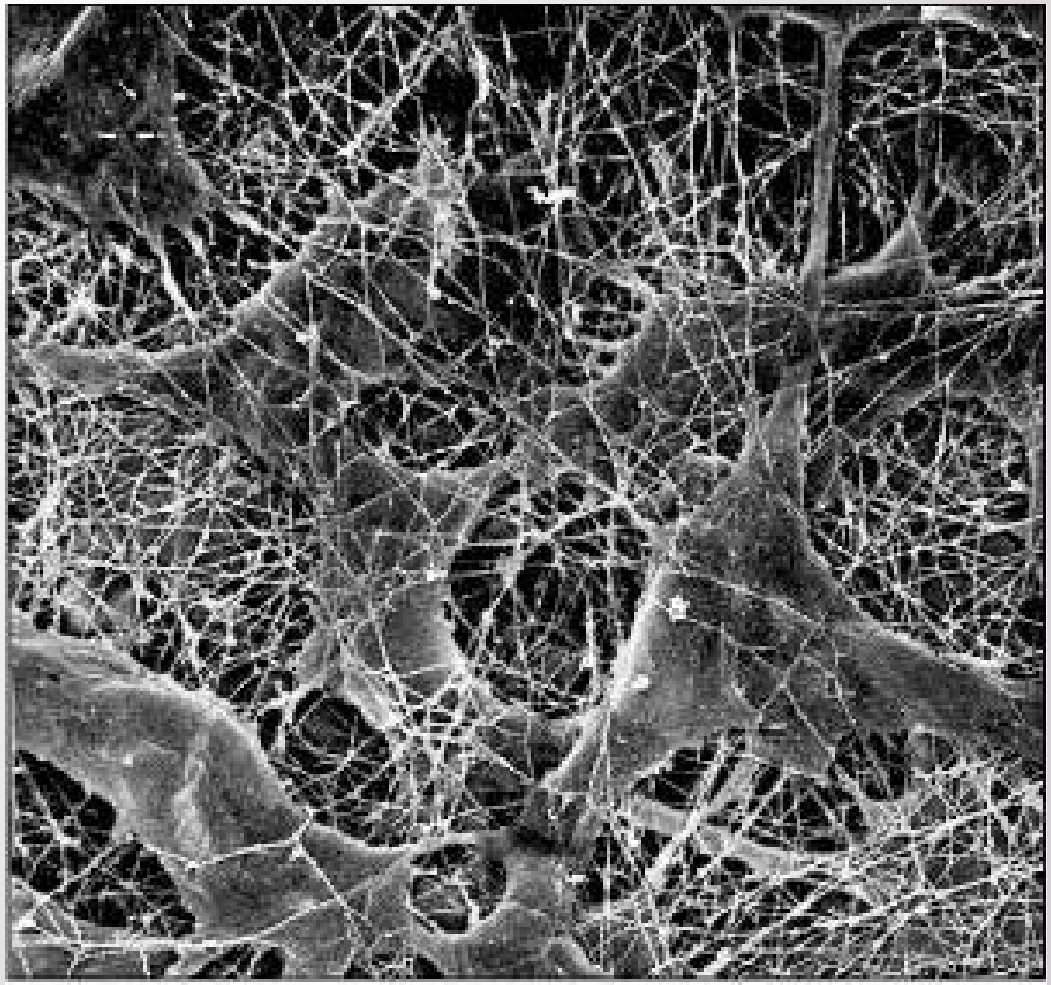
**Tight junctions block lateral movement of lipids and membrane proteins to keep a cell polarized. They leave no space between plasma membranes of adjacent cells to prevent the movement of molecules across cell layers. Sodium/glucose symport proteins and export by glucose transport proteins on the basolateral surface and tight junctions prevent the lateral movement of these transport proteins.**

# Summary of cell junctions found in the cell

| name | function |
| --- | --- |
| tight junction | seals neighboring cells together in an epithelial sheet to prevent leakage of molecules between them |
| adherens junction | joins an actin bundle in one cell to a similar bundle in a neighboring cell |
| desmosome | joins the intermediate filaments in one cell to those in a neighbor |
| gap junction | allows the passage of small water-soluble ions and molecules |
| hemidesmosome | anchors intermediate filaments in a cell to the basal lamina |

Figure 19–19 part 2 of 2. Molecular Biology of the Cell, 4th Edition.

# Fibroblasts surrounded by collagen fibrils



There are 3 types of students.
1)some make wonders happen
2)some watch wonders happen
3)some wonders,what happen?
:p

- Donno



boardofwisdom.com

# Fibroblasts

- are the predominant cells of connective tissue.
- They are responsible for the formation and maintenance of the fibrous components and the ground substance of connective tissue.